

BIRJAND, Iran, Islamic Republic of

WMO: 408090

Lat: 32.8907N

Lon: 59.2839E

Elev: 4892

StdP: 12.28

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.5	21.2	-11.5	3.5	41.2	-5.5	4.9	44.7	24.1	47.4	21.1	47.7	3.7	120	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.6	100.8	61.2	98.6	60.3	96.4	59.6	64.9	95.9	63.1	93.7	61.4	91.2	10.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
52.1	69.4	73.0	50.0	64.1	73.6	48.1	59.7	72.3	32.6	96.4	31.1	93.6	29.8	91.3	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	19.6	17.2	DB	9.3	105.0	5.2	2.0	5.6	106.4	2.5	107.5	-0.4	108.6	-4.2	110.1	(4)
(5)			WB	7.2	65.9	4.6	3.6	3.9	68.5	1.2	70.6	-1.5	72.6	-4.8	75.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.2	39.8	44.3	53.6	63.3	72.8	80.3	82.8	78.6	72.3	63.2	51.0	43.0		
	DBStd	16.12	7.58	8.06	7.62	6.98	5.96	5.02	4.23	4.46	5.17	6.30	7.11	6.80		
	HDD50	876	325	190	51	4	0	0	0	0	0	0	73	233		
	HDD65	3058	782	579	358	110	9	0	0	0	7	109	422	682		
	CDD50	5320	8	31	163	403	707	908	1017	888	668	409	102	16		
	CDD65	2027	0	0	5	59	251	458	552	423	225	53	1	0		
	CDH74	28369	0	6	114	818	3345	6340	7519	5677	3436	1071	42	1		
	CDH80	14836	0	0	23	202	1545	3627	4369	3085	1673	310	2	0		
(14) Wind	WSAvg	6.9	5.3	6.0	6.9	7.0	7.5	8.7	10.0	8.6	6.6	5.6	4.9	5.1		
(15) Precipitation	PrecAvg	6.8	1.4	1.5	1.6	0.7	0.3	0.0	0.0	0.0	0.1	0.1	0.3	0.8		
	PrecMax	10.9	2.9	2.8	5.6	2.1	1.3	0.3	0.0	0.5	1.4	0.9	1.1	3.1		
	PrecMin	2.8	0.4	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.1	0.6	0.9	1.2	0.6	0.3	0.1	0.0	0.1	0.3	0.2	0.3	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.0	73.6	83.7	89.3	97.1	103.9	104.3	102.9	96.8	90.0	78.8	71.0		
		MCWB	45.0	47.9	52.3	54.8	58.8	61.9	63.5	63.6	57.6	55.6	53.1	47.7		
	2%	DB	62.3	67.0	76.8	85.3	93.9	100.6	101.8	98.5	93.9	86.4	74.1	64.8		
		MCWB	43.8	46.6	50.4	55.0	58.0	61.0	61.8	60.5	56.9	55.2	50.0	45.1		
	5%	DB	58.7	62.8	72.9	82.2	91.4	98.3	98.8	95.3	91.7	83.4	70.4	61.2		
		MCWB	42.4	45.0	49.0	53.8	57.3	59.8	60.9	59.2	56.3	54.0	48.5	43.3		
	10%	DB	54.5	59.2	69.5	79.0	88.2	95.2	96.4	92.9	89.4	80.3	67.0	57.6		
		MCWB	40.6	43.3	48.1	52.4	56.6	58.8	60.3	58.2	55.9	52.6	47.0	42.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	51.5	55.2	58.4	61.9	65.3	68.0	66.3	62.5	59.0	55.4	49.9		
		MCDB	57.0	64.0	69.6	81.2	91.2	99.7	97.8	96.2	91.2	83.9	72.9	62.2		
	2%	WB	46.5	49.1	53.1	56.8	60.2	63.2	65.7	63.5	59.7	56.8	52.3	47.4		
		MCDB	57.0	61.2	68.7	79.1	88.7	96.4	96.1	93.4	88.7	81.9	67.5	60.0		
	5%	WB	44.4	47.4	51.3	55.4	58.8	61.5	63.9	61.6	58.1	55.0	50.2	45.5		
		MCDB	56.1	58.9	67.0	76.3	86.4	93.4	94.2	91.0	86.9	79.9	66.6	57.7		
	10%	WB	41.9	45.6	49.6	54.0	57.5	59.9	62.0	59.9	56.5	53.4	48.4	43.7		
		MCDB	52.2	56.6	65.2	73.7	84.2	91.1	91.5	88.9	85.6	77.7	64.5	55.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	24.5	23.7	24.4	25.2	26.6	27.1	25.6	28.2	32.0	31.3	28.0	26.8		
		MCDBR	28.5	27.5	28.5	28.4	29.3	29.5	29.0	30.6	33.7	32.3	31.0	30.7		
		MCWBR	15.8	13.9	12.5	11.8	11.3	11.2	10.7	12.0	14.5	15.0	16.1	17.3		
		MCDBR	23.5	21.7	22.8	24.8	26.5	27.5	26.0	27.3	30.8	30.4	26.6	25.0		
	5% WB	MCWBR	13.9	11.7	10.9	10.5	10.8	11.4	11.1	11.7	14.4	15.0	14.4	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.365	0.441	0.506	0.518	0.479	0.452	0.413	0.375	0.384	0.348	0.306		
	taud		2.341	2.157	1.926	1.802	1.788	1.884	1.966	2.073	2.161	2.117	2.244	2.385		
	Ebn at Noon		285	278	263	252	250	259	265	274	281	267	268	282		
	Edh at Noon		32	42	58	68	70	63	58	51	45	43	35	29		
(44) All-Sky Solar Radiation	RadAvg		1036	1290	1679	2004	2298	2630	2593	2434	2082	1615	1185	1000		
	RadStd		105	107	97	118	112	82	40	60	47	56	69	83		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+0.76	N/A	-6.30	+1.28	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (2 neighbors)	+0.76	N/A	-6.30	+1.28	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page